

Quadratics Assimilation v2 - Benchmark

- Target learner: about 50% prior knowledge. The student may remember formulas but has unstable connections, recognition, method choice, and transfer.
- Macro loop: SEE -> REALIZE -> UNDERSTAND -> ADOPT.
- Operational loop: RECONNECT -> DISCOVER -> MAKE SENSE -> TRY -> DIAGNOSE -> FADE -> ADOPT -> TRANSFER.
- Central rule: the requested information chooses the representation.
- Benchmark use: compare pedagogy, completeness, source discipline, and production quality. Do not copy wording or layout.

1. Representations and method choice

- RECONNECT with $x^2-5x+6 = 0$: standard form, factor form, roots 2 and 3, graph intercepts.
- DISCOVER that no single representation is always best.
- Contrast: find larger root vs find $\alpha^2+\beta^2$ for the same quadratic. First asks for individual roots; second is symmetric and should trigger Vieta.
- DIAGNOSE: automatic quadratic-formula use creates information the question may not need.
- ADOPT: classify the target before calculating.

2. Discriminant as root geometry

- RECONNECT: $D=b^2-4ac$ is remembered, but meaning may be weak.
- MAKE SENSE through $x=(-b \pm \sqrt{D})/(2a)$: $D>0$ gives two real roots, $D=0$ a repeated root, $D<0$ no real roots.
- Graph bridge: repeated root means tangency to the x-axis.
- Contrast: equal roots $\rightarrow D=0$; minimum value $\rightarrow$ complete square / vertex. Similar-looking parameter questions require different structures.
- FADING: H3 equation, H2 structural cue, H1 recognition cue, then H0.

3. Vieta and symmetric targets

- Derive from $a(x-\alpha)(x-\beta)$, not from memorization.
- $S=\alpha+\beta=-b/a$ and $P=\alpha\beta=c/a$.
- Symmetry test: if swapping α and β does not change the target, first try S and P .
- Examples: $\alpha^2+\beta^2=S^2-2P$; $(\alpha-\beta)^2=S^2-4P$; $1/\alpha+1/\beta=S/P$.
- Contrast: larger root needs individual-root information; symmetric expressions usually do not.

4. Transformed and integer roots

- Transformed roots: shifted, reciprocal, squared. Compute the new sum/product before solving original roots.
- Decision boundary: transformed roots are not the same as shifting a function input.
- Positive product means same sign, not automatically positive roots; use the sum too.
- Positive integer roots add discrete factor-pair/divisibility structure.
- Source conflicts remain explicit; never change a sign/key silently to make an answer fit.

5. Quadratic relation as rewriting machine

- If $x^2=x+1$ and the target is x^{20} , solving roots may be wasteful.
- Repeated substitution gives $x^3=2x+1$, $x^4=3x+2$, $x^5=5x+3$: a recurrence appears.
- Invariant: a quadratic relation reduces every higher power to $ax+b$.
- Contrast: find roots of $x^2-x-1=0$ vs simplify x^8 under $x^2=x+1$.
- Teach transform/reduce before calculate when the surface is large but the structure is small.

6. Vertex view and parameter translation

- Complete the square to expose minimum/maximum value: $x^2-6x+5=(x-3)^2-4$.
- Phrase decoder: equal roots $\rightarrow D=0$; distinct real roots $\rightarrow D>0$; no real roots $\rightarrow D<0$;
positive roots $\rightarrow$ real plus $S>0$ and $P>0$; prescribed minimum/maximum $\rightarrow$ vertex condition.
- The phrase should be translated to a mathematical condition before arithmetic.
- Contrast root-count language with value-of-expression language.

7. Practice architecture and fading

- TRY before full solution. Prefer first-move-only prompts.
- DIAGNOSE why the wrong method is tempting and identify the missing bridge: recognition, representation, invariant, domain, execution, or checking.
- FADE H3 -> H2 -> H1 -> H0. Do not repeat worked scaffolding indefinitely.
- First-Step Reference is built only after concept teaching. It compresses recognition atlas, phrase decoder, decision tree, contrast pairs, traps, and first moves.

8. Mastery, transfer, and source custody

- Six-question assimilation test: What did you notice? Why does it work? What clue triggers it? What near-miss needs a different method? Can you write first two lines? Can you solve a disguised version?
- Mastery paper should be H0 and unlabelled by method.
- Transfer must change surface/representation, not only numbers.
- Independently recompute answers before promotion. A real audit caught $(\alpha - \beta)^2 = 33/4$ for $2x^2 + x - 4 = 0$.
- Static QA can pass; classroom timing/readability and longitudinal mastery remain NOT_RUN until observed.